

EX 14.2.5

$$\begin{array}{c}
 P1 \\
 \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array}
 \end{array}
 \begin{array}{c}
 P2 \\
 \begin{array}{c} b_1 \\ b_2 \\ b_3 \\ b_4 \end{array}
 \end{array}
 \left[\begin{array}{cccc}
 3 & -3 & -2 & -4 \\
 -4 & -2 & -1 & 1 \\
 1 & \boxed{-1} & 2 & 0
 \end{array} \right]$$

$a_1 \rightarrow b_4 \rightarrow a_2 \rightarrow b_1 \rightarrow a_1$ NOPE

a_2 (same as above)

$a_3 \rightarrow b_2 \rightarrow a_3$

$$R_1(b_2) = a_3$$

$$R_2(a_3) = b_2$$

ALTERNATIVE

$$\begin{array}{c}
 \text{max col}
 \end{array}
 \left[\begin{array}{cccc}
 3 & -3 & -2 & -4 \\
 -4 & -2 & -1 & 1 \\
 1 & \boxed{-1} & 2 & 0 \\
 3 & \boxed{-1} & 2 & 0
 \end{array} \right]
 \begin{array}{c}
 \text{min row} \\
 -4 \\
 -4 \\
 \boxed{-1}
 \end{array}$$

EX 14.2.3

P_1 max P_2 min

	1	2	3	4	
1	5	-7	-2	2	$\left. \begin{array}{l} \rightarrow 3. \text{ dominated by } 3 \\ \rightarrow 4. \text{ dominated by } 3 \end{array} \right\}$
2	-2	2	-5	5	
3	-2	5	-2	7	

1. dominated by 3 2. dominated by 3
 3. dominated by 3
 4. dominated by 3

P_1 P_2

	a_1	a_2	a_3	a_4
a_1	5	-7	-2	2
a_2	-2	2	-5	5
a_3	-2	5	-2	7
b_1		b_2	b_3	b_4

$a_1 \rightarrow b_2 \rightarrow a_3 \rightarrow b_3 \rightarrow a_3 \rightarrow b_3 \rightarrow a_1$
 (UNSTABLE)

best response set
 $R_1(b_3) = a_3$ $R_2(a_3) = b_3$
 (UNSTABLE)

EXERCISE

		L	C	R	
P1	T	2, 1	3, 4	5, 3	p
	B	3, 4	1, 1	1, 2	$(1-p)$

if something, fine strategy

PROFIT: $T \rightarrow C \rightarrow T$
 $B \rightarrow L \rightarrow B$
 $R \rightarrow T \rightarrow C \rightarrow T$

PURE EQ (T, C), (B, L)

COST: $T \rightarrow L \rightarrow T$
 $B \rightarrow C \rightarrow B$
 $R \rightarrow B \rightarrow C$

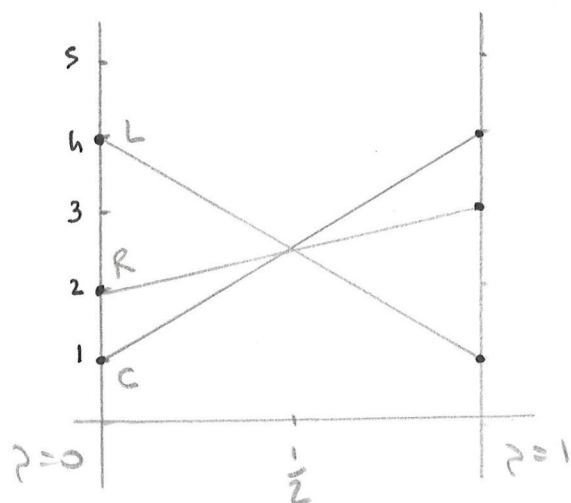
PURE EQ (T, L), (B, C)

MIXED EQ:

PROFIT: EXPECTED PAYOFF

$$\begin{array}{ll} P1 & 2q_1 + 3q_2 + 5q_3 \quad (T) \\ & 3q_1 + q_2 + q_3 \quad (B) \end{array}$$

$$\begin{array}{ll} P2 & p + 4(1-p) \quad (L) \\ & 4p + (1-p) \quad (C) \\ & 3p + 2(1-p) \quad (R) \end{array}$$



$$p + 4(1-p) = 3p + 2(1-p)$$

$$p + 4 - 4p = 3p + 2 - 2p$$

$$-3p + 4 = p + 2$$

$$-4p = -2$$

$$p = \frac{1}{2}$$

$$V = \frac{5}{2}$$

P1 $2q_1 + 3q_2 + 5q_3 \geq 3q_1 + q_2 + q_3$ if (T) gives
a higher payoff than (B)

$2q_1 + 3q_2 + 5q_3 \leq 3q_1 + q_2 + q_3$ if (B) is better

$$2q_1 + 3q_2 + 5q_3 = 3q_1 + q_2 + q_3$$

$$\Rightarrow q_1 = 2q_2 + 4q_3 \quad (\text{infinite possibilities})$$

$$\text{if } q_2 = 0$$

$$q_1 = 4q_3$$

$$q_1 = \frac{4}{5} \quad q_3 = \frac{1}{5} \quad (q_1 + q_2 + q_3 = 1)$$

$$\text{if } q_3 = 0$$

$$q_1 = 2q_2$$

$$q_1 = \frac{2}{3} \quad q_2 = \frac{1}{3}$$

$$\text{if } q_1 = 0$$

$$2q_2 = -4q_3 \quad \text{cannot happen}$$

$$V = 2 \cdot \frac{2}{3} + 3 \cdot \frac{1}{3} + 5 \cdot 0$$

$$= \frac{4}{3} + 1 = \frac{7}{3}$$

EXERCISE

FIND ALL NASH EQUILIBRIA

$$1. \begin{bmatrix} 5, 2 & 1, 1 \\ 0, 2 & 4, 3 \\ 3, 0 & 4, 4 \end{bmatrix} \quad \text{PROFIT}$$

$$2. \begin{bmatrix} 5, 3 & 1, 1 \\ 0, 2 & 5, 3 \\ 3, 0 & 4, 4 \end{bmatrix} \quad \text{PROFIT}$$

$$3. \begin{bmatrix} 5, 2 & 1, 1 \\ 0, 2 & 4, 3 \\ 2, 0 & 2, 5 \end{bmatrix} \quad \text{PROFIT}$$

$$4. \begin{bmatrix} 2, 4 & 3, 2 & 1, 0 \\ 0, 1 & 2, 2 & 3, 3 \end{bmatrix} \quad \text{PROFIT}$$

$$\begin{array}{cc}
 & P_2 \\
 \begin{array}{c} I. \\ P_1 \end{array} & \begin{array}{cc} \boxed{5, 2} & 1, 1 \\ 0, 2 & \boxed{4, 3} \\ 3, 0 & 4, 4 \end{array} \\
 & \begin{array}{c} q \quad 1-q \end{array} \\
 & \begin{array}{c} T_1 \\ T_2 \\ T_3 \end{array}
 \end{array}$$

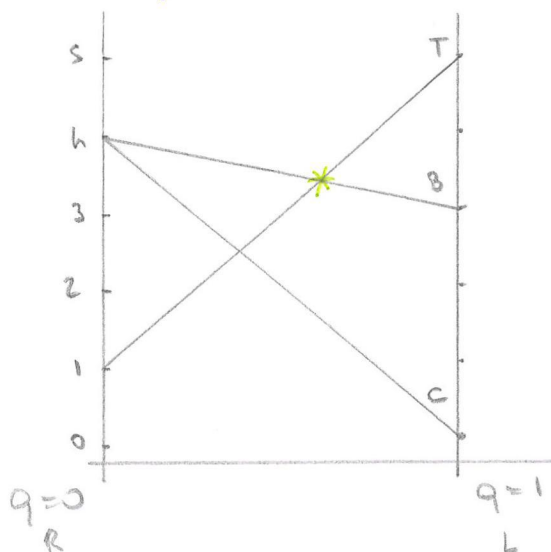
BEST RESPONSE:

$$\begin{aligned}
 R_1(L) &= T \\
 R_1(R) &= \{C, B\} \\
 R_2(T) &= L \\
 R_2(C) &= R \\
 R_2(L) &= R
 \end{aligned}$$

PURE STRATEGY EQ: (T, L) (C, R)

HYBRID STRATEGY:

P_1	$5q + 1(1-q)$	T
	$0q + 4(1-q)$	C
	$3q + 4(1-q)$	B



$$\begin{aligned}
 * \quad 5q + 1 - q &= 3q + 4(1 - q) \\
 4q + 1 &= -q + 4 \\
 q &= \frac{3}{5} \quad v = \frac{17}{5}
 \end{aligned}$$

P_2 : since P_1 is indifferent to C, P_2 should not play it
 $\Rightarrow P_2 = 0$
 $\downarrow$
 "dominated by B"

$$\begin{array}{ll} P2: & 2p_1 \quad (L) \\ & p_1 + 3p_2 + 4p_3 \quad (R) \end{array}$$

$$2p_1 = p_1 + 4p_3 \quad (p_2 = 0)$$

$$p_1 = 4p_3$$

$$p_1 = \frac{4}{5} \quad p_3 = \frac{1}{5} \quad (p_1 + p_2 + p_3 = 1)$$

$$v = \frac{8}{5}$$

NASH EQ:

- $p = (1, 0, 0) \quad q = (1, 0) \quad v = (5, 2)$
- $p = (0, 1, 0) \quad q = (0, 1) \quad v = (4, 3)$
- $p = (\frac{4}{5}, 0, \frac{1}{5}) \quad q = (\frac{2}{5}, \frac{3}{5}) \quad v = (\frac{17}{5}, \frac{8}{5})$

2.

		P2		
		T	C	
P1	L	5, 3	1, 1	P1
	B	0, 2	5, 3	P2
	R	3, 0	4, 4	P3
		1-q	q	

BEST RESPONSE: $R_1(L) = T$

$$R_1(R) = C$$

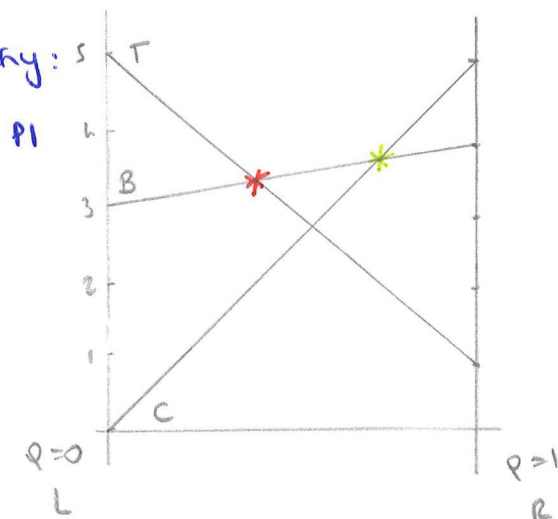
$$R_2(T) = L$$

$$R_2(C) = R$$

$$R_2(B) = R$$

PURE EQ: (T, L) (C, R)

MIXED STRATEGY:



$$5(1-q) + q \quad (T)$$

$$0(1-q) + 5q \quad (C)$$

$$3(1-q) + 4q \quad (B)$$

$$5q = 3(1-q) + 4q \quad \dots$$

$$5q = 3 - 3q + 4q$$

$$4q = 3$$

$$q = \frac{3}{4} \quad v = \frac{15}{4}$$

$$P2 \quad 3p_1 + 2p_2 + 0p_3 \quad (L)$$

$$p_1 + 3p_2 + 4p_3 \quad (R)$$

$$p_1 = 0 \quad (\text{indifferent for } P1)$$

$$2p_2 = 3p_2 + 4p_3$$

$$-p_2 = 4p_3 \quad \text{IMPOSSIBLE!}$$

Both the players should be willing to play the mixed strategy

let's check *

$$5(1-q) + q = 3(1-q) + 4q$$

$$5 - 5q + q = 3 - 3q + 4q$$

$$5 - 4q = 3 + q$$

$$-5q = -2$$

$$q = \frac{2}{5} \quad v = \frac{17}{5}$$

↳ new mixed strategy

$$P2: \quad 3p_1 + 2p_2 + 0p_3 \quad (L)$$

$$p_1 + 3p_2 + 4p_3 \quad (R)$$

$$p_2 = 0 \quad (C \text{ indifferent})$$

$$3p_1 = p_1 + 4p_3$$

$$2p_1 = 4p_3 \quad p_1 = 2p_3$$

$$p_1 = \frac{2}{3} \quad p_3 = \frac{1}{3} \quad (p_1 + p_2 + p_3 = 1)$$

3.
$$P_1 \begin{matrix} & \begin{matrix} P_2 \\ L & R \end{matrix} \\ \begin{matrix} S \\ C \\ B \end{matrix} & \begin{bmatrix} \boxed{5, 2} & 1, 1 \\ 0, 2 & \boxed{4, 3} \\ 2, 0 & 2, 5 \end{bmatrix} \end{matrix}$$

BEST RESPONSE: $R_1(L) = T$

$R_1(R) = C$

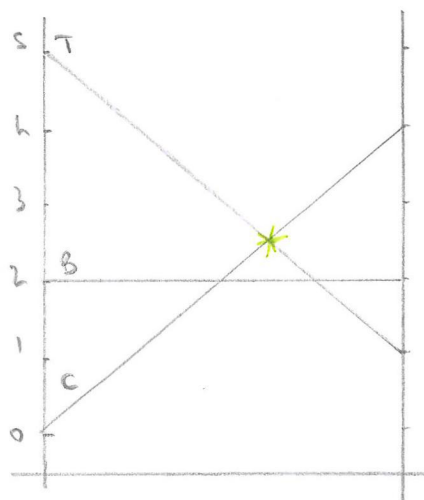
$R_2(T) = L$

$R_2(C) = R$

$R_2(B) = R$

PURE EQ: (T, L) (C, R)

MIXED STRATEGY: P_1



$$5(1-q) + q \quad (T)$$

$$0(1-q) + 4q \quad (L)$$

$$2(1-q) + 2q \quad (B)$$

$$* \quad 5(1-q) + q = 4q$$

$$-3q = -5$$

$$q = \frac{5}{3} \quad V = \frac{5}{2}$$

$$P_2: \quad p_3 = 0$$

$$2p_1 + 2p_2 \quad (L)$$

$$p_1 + 3p_2 \quad (R)$$

$$2p_1 + 2p_2 = p_1 + 3p_2$$

$$p_1 = p_2 \quad p_1 = \frac{1}{2} \quad p_2 = \frac{1}{2} \quad V = 2$$

4.
$$\begin{bmatrix} \boxed{2,4} & 3,2 & 1,0 \\ 0,1 & 2,2 & \boxed{3,3} \end{bmatrix} \begin{matrix} 1-p \\ p \end{matrix}$$

BEST RESPONSE:

$$R_1(L) = T$$

$$R_1(C) = T$$

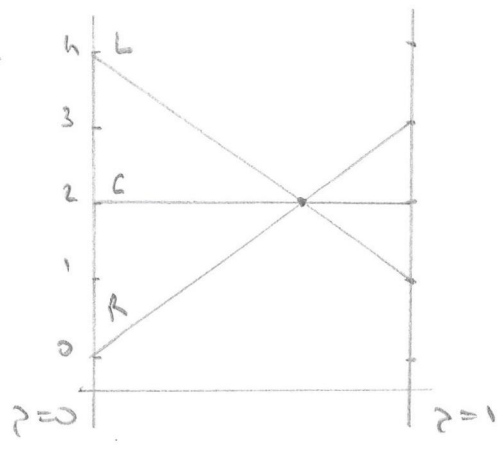
$$R_1(R) = B$$

$$R_2(T) = L$$

$$R_2(B) = R$$

PURE EQ: (T, L) (B, R)

MIXED EQ: P2



$$\begin{aligned} 4(1-p) + p & \quad (L) \\ 2(1-p) + 2p & \quad (C) \\ 0(1-p) + 3p & \quad (R) \end{aligned}$$

$$2(1-p) + 2p = 3p$$

$$2 - 2p + 2p = 3p$$

$$2 = 3p$$

$$p = \frac{2}{3}$$

P1

$$2q_1 + 3q_2 + q_3 \quad (T)$$

$$0q_1 + 2q_2 + 3q_3 \quad (B)$$

$$2q_1 + 3q_2 + q_3 = 2q_2 + 3q_3$$

$$2q_1 + q_2 - 2q_3 = 0$$

$$q_2 = 2q_3 - 2q_1$$

INFINITE POSSIBILITIES!

$$q_1 = 0 \quad q_2 = \frac{2}{3} \quad q_3 = \frac{1}{3}$$

$$q_2 = 0 \quad q_1 = \frac{1}{2} \quad q_3 = \frac{1}{2}$$